King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam I Term 232

Original exam date: 18 February 2024. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. $\sum_{i=1}^{7} i(i+3)^2 =$

(a) 1786

(b) 1678

(c) 1787

(d) 1876

(e) 1668

2. $\int x\sqrt{x+6}\,dx =$

(a) $\frac{3}{2}(x-6)^{1/2}(x-1)^{3/2}+C$

(b) $\frac{2}{3}(x+6)^{2/3}(x-4)+C$

(c) $\frac{4}{5}(x+6)^{3/2}(x+16)+C$

(d) $\frac{2}{5}(x-4)^{3/2}(x+6)+C$

(e) $\frac{2}{5}(x+6)^{3/2}(x-4)+C$

3. $\int_{-\pi/4}^{\pi/4} \left(x^3+x^4\tan x\right)dx =$

(a) $-\frac{\pi}{2}$

(b) $\frac{\pi}{2}$

(c) 0

(d) $\left(\frac{\pi}{4}\right)^4 \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(e) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

4. $\int_0^{\pi/2} \cos x \, \sin(\sin x)\, dx =$

(a) -1

(b) $1 - \cos 1$

(c) 0

(d) 1

(e) $\pi - 1$

5. Given that $\int_2^6 x^3\, dx = A$ and $\int_2^6 x\, dx = B$, the integral $\int_2^6 \left(6|x| - \frac{1}{8}x^3\right) dx =$

(a) $6B - \dfrac{1}{8}A$

(b) $12B - \dfrac{1}{8}A$

(c) $2B - \dfrac{1}{8}A$

(d) $-6B - \dfrac{1}{8}A$

(e) $-12B + \dfrac{1}{8}A$

6. If A is the exact area of the region bounded between the two curves $f(x) = \cos x$ and $g(x) = 2 - \cos x$ over the interval $[0, 2\pi]$, then

(a) $5 < A < 6$

(b) $10 < A < 11$

(c) $12 < A < 13$

(d) $9 < A < 10$

(e) $7 < A < 8$

7. When evaluating the limit of $\sum_{i=1}^{n} \left(1 + \frac{3}{c_i}\right) \Delta x$, as $\|\Delta x\| \to 0$ over $[1, 5]$, where c_i is any point in the ith subinterval, the value is equal to:

(a) $4\ln(5)$

(b) $4 + \ln(125)$

(c) $\dfrac{3}{5}$

(d) $\dfrac{8}{5}$

(e) $\dfrac{-12}{5}$

8. Using the midpoint rule with $n = 3$, the approximate area of the region bounded by the graph of $f(x) = \sin \pi x$ and the x-axis over $\left[0, \frac{3}{2}\right]$ is equal to

(a) $\frac{\sqrt{2}}{4}$

(b) $\frac{3}{4}\sqrt{2}$

(c) 0

(d) $\frac{3\sqrt{2}}{2}$

(e) $\frac{\sqrt{2}}{2}$

9. The area of the region bounded between $y_1 = (x-1)^3$ and $y_2 = x - 1$ is equal to

(a) 1

(b) $\frac{1}{2}\sqrt[3]{2}$

(c) $2\sqrt[3]{2}$

(d) $\frac{1}{4}$

(e) $\frac{1}{2}$

10. $\lim_{n\to\infty} \frac{\sqrt{1}+\sqrt{2}+\sqrt{3}+\cdots+\sqrt{n}}{n^{3/2}}$ equals to

(a) $\sqrt{3}$

(b) $\frac{3}{2}$

(c) ∞

(d) 0

(e) $\frac{2}{3}$

11. $\int \frac{x^4+x-4}{x^2+2}\,dx =$

(a) $\frac{1}{2}x^2 - x + 2\ln\sqrt{x^2+2} + C$

(b) $\frac{2}{3}x^3 + \ln|x| + \ln(x^2+2) + C$

(c) $\frac{1}{3}x^3 - 2x + \ln\sqrt{x^2+2} + C$

(d) $\frac{1}{3}x^3 + 2x + \ln(x^2-4) + C$

(e) $\frac{1}{3}x^3 + \ln|x| + \ln(x^2-4) + C$

12. $\int_9^1 \frac{\sqrt{2}}{\sqrt{x}\,(1+\sqrt{x})^2}\,dx =$

(a) $\dfrac{-1}{2}$

(b) $8\sqrt{2}$

(c) $\dfrac{1}{2}$

(d) $\dfrac{-\sqrt{2}}{2}$

(e) $\dfrac{\sqrt{2}}{2}$

13. $\int \frac{x}{x^4+25}\,dx =$

(a) $\dfrac{1}{25}\tan^{-1}\left(\dfrac{x^2}{25}\right)+C$

(b) $\dfrac{1}{5}\tan^{-1}\left(\dfrac{x^2}{10}\right)+C$

(c) $\dfrac{1}{5}\tan^{-1}\left(\dfrac{x}{10}\right)+C$

(d) $\dfrac{1}{10}\tan^{-1}\left(\dfrac{x}{5}\right)+C$

(e) $\dfrac{1}{10}\tan^{-1}\left(\dfrac{x^2}{5}\right)+C$

14. The constant c that is guaranteed by the mean value theorem of integrals for $f(x) = 5 - \dfrac{1}{x}$ over $[1, 4]$ is equal to

(a) $\dfrac{3}{\ln 4}$

(b) $\dfrac{1}{2\ln 2}$

(c) $\dfrac{1}{2\ln 2 + 10}$

(d) $\dfrac{1}{\ln 2}$

(e) $\dfrac{1}{\ln 4 - 10}$

15. $\int_0^{\ln 2} 2e^{-x}\cosh x\,dx =$

(a) $2\ln 2 + e$

(b) $\dfrac{1+\ln 2}{e}$

(c) $e^{-1}\ln 2$

(d) $\dfrac{2+\ln 2}{3}$

(e) $\dfrac{3}{8} + \ln 2$

16. $\int_{\pi/8}^{\pi/4} (\csc 2\theta - \cot 2\theta)\,d\theta =$

(a) $\dfrac{1}{2}\ln\left(\dfrac{\sqrt{2}+\pi}{\sqrt{2}-\pi}\right)$

(b) $\dfrac{1}{4}\ln(\pi - 1)$

(c) $\dfrac{\sqrt{2}}{2}\ln(\pi + 1)$

(d) $\ln\sqrt{1+\dfrac{\sqrt{2}}{2}}$

(e) $\dfrac{1}{2}\ln(\sqrt{2}-1)$

17. $\int \frac{1}{x\sqrt{x^4-4}}\,dx =$

(a) $\dfrac{1}{2}\sec^{-1}\left(\dfrac{x^2}{4}\right) + C$

(b) $\dfrac{1}{4}\sin^{-1}\left(\dfrac{x^2}{2}\right) + C$

(c) $\dfrac{1}{4}\sec^{-1}\left(\dfrac{x^2}{2}\right) + C$

(d) $\dfrac{1}{2}\sin^{-1}\left(\dfrac{x^2}{4}\right) + C$

(e) $\dfrac{1}{2}\sin^{-1}\left(\dfrac{x^2}{2}\right) + C$

18. If $F(x) = \int_0^{2x} \cos t^4 \, dt$, then $F'(x) =$

(a) $32\cos(x^4)$

(b) $\cos(16x^4)$

(c) $-2\cos(16x^4)$

(d) $2\cos(16x^4)$

(e) $-\cos(16x^4)$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$i^3 + 6i^2 + 9i$: $784 + 840 + 252 = 1876$.
2	**(e)**	$u = x + 6$: $\frac{2}{5}u^{5/2} - 4u^{3/2} = \frac{2}{5}(x+6)^{3/2}(x-4) + C$.
3	**(c)**	Both terms are odd on a symmetric interval, so the integral is 0.
4	**(b)**	$u = \sin x$: $\int_0^1 \sin u\, du = 1 - \cos 1$.
5	**(a)**	$\lvert x\rvert = x$ on $[2,6]$, so the integral is $6B - \frac{1}{8}A$.
6	**(c)**	$A = \int_0^{2\pi}(2 - 2\cos x)dx = 4\pi \approx 12.57$.
7	**(b)**	$\int_1^5(1 + \frac{3}{x})dx = 4 + 3\ln 5 = 4 + \ln 125$.
8	**(b)**	$\Delta x = \frac{1}{2}$, midpoints $\frac{1}{4}, \frac{3}{4}, \frac{5}{4}$: $\frac{1}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{4}$.
9	**(e)**	Intersections at $x = 0, 1, 2$; $A = 2\int_0^1(u - u^3)du = \frac{1}{2}$.
10	**(e)**	Riemann sum of $\sqrt{x}$ on $[0,1]$: $\int_0^1 \sqrt{x}\, dx = \frac{2}{3}$.
11	**(c)**	Divide: $x^2 - 2 + \frac{x}{x^2+2}$, giving $\frac{1}{3}x^3 - 2x + \ln\sqrt{x^2+2} + C$.
12	**(d)**	$u = 1 + \sqrt{x}$: $2\sqrt{2}\int_4^2 u^{-2}du = -\frac{\sqrt{2}}{2}$.
13	**(e)**	$u = x^2$: $\frac{1}{2}\int \frac{du}{u^2+25} = \frac{1}{10}\tan^{-1}\frac{x^2}{5} + C$.
14	**(a)**	Average $= 5 - \frac{\ln 4}{3}$; $5 - \frac{1}{c}$ equal to it gives $c = \frac{3}{\ln 4}$.
15	**(e)**	$2e^{-x}\cosh x = 1 + e^{-2x}$; integral $= \ln 2 + \frac{3}{8}$.
16	**(d)**	$\csc 2\theta - \cot 2\theta = \tan\theta$; result $\frac{1}{2}\ln\left(1 + \frac{\sqrt{2}}{2}\right)$.
17	**(c)**	$u = x^2$: $\frac{1}{2}\int \frac{du}{u\sqrt{u^2-4}} = \frac{1}{4}\sec^{-1}\frac{x^2}{2} + C$.
18	**(d)**	$F'(x) = \cos((2x)^4) \cdot 2 = 2\cos(16x^4)$.